Data Analyst Job Postings: Salary, Experience, and Programming Language Requirements

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1. Introduction

This report analyzes a sample of 400 Data Analyst job postings to explore how salary estimates vary by minimum years of experience and by programming language requirements. The aim is to use simple visual analysis to identify patterns in the job market while recognizing that the results are based on a limited sample and should not be generalized to all Data Analyst roles.

2. Analysis of Years of Experience

The first visualization is a scatter plot showing Minimum Years Experience on the x-axis and Median Salary Estimate on the y-axis. Records with non-numeric or missing experience values were excluded from this chart, leaving 311 usable observations.

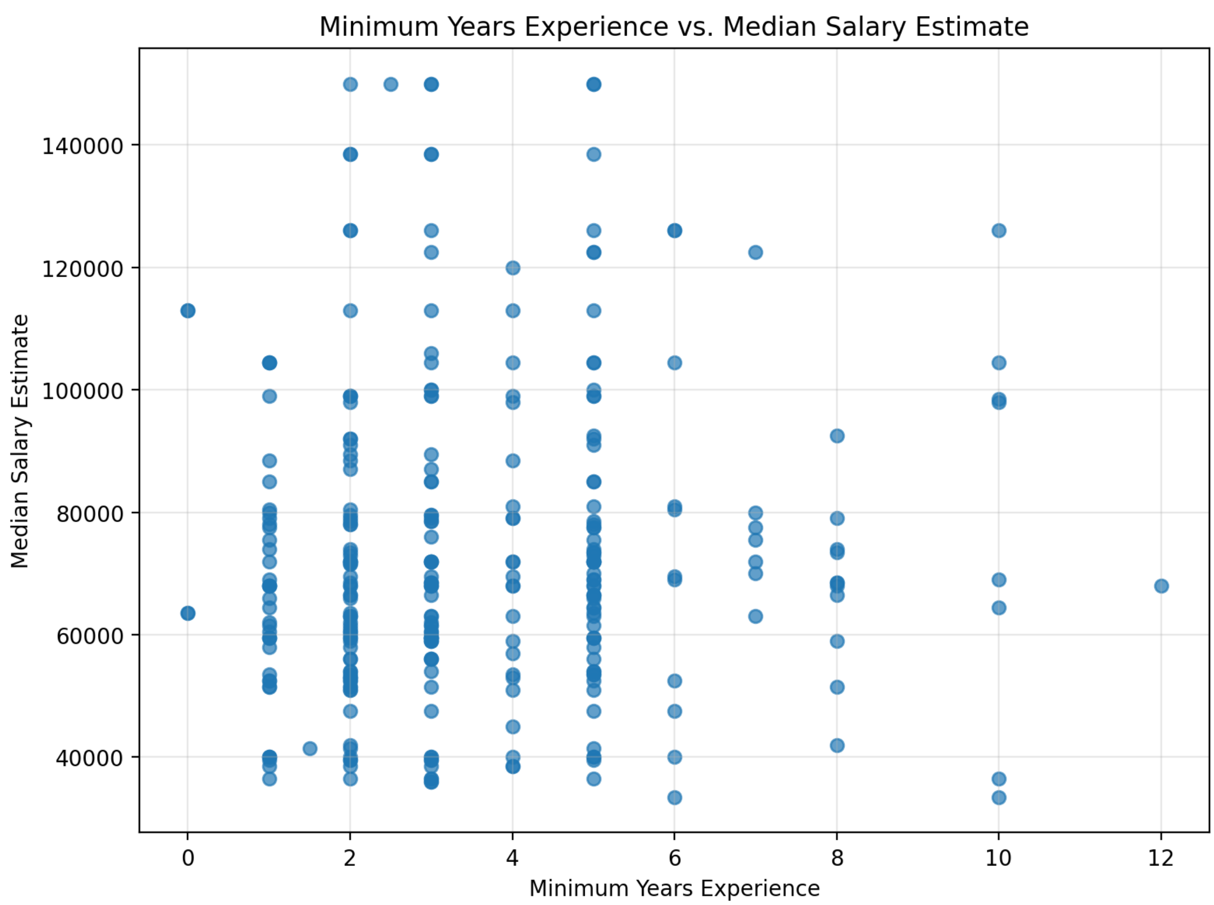

Figure 1. Minimum Years Experience vs. Median Salary Estimate

Interpretation. The scatter plot shows a wide range of salary estimates across most experience levels. While higher salaries appear at several experience levels, the relationship between years of experience and salary is not strongly linear in this sample. The calculated correlation is very weak, suggesting that salary is likely influenced by other factors such as location, company size, industry, job responsibilities, seniority level, and the quality of the extracted posting data. A key limitation is that only 311 postings had valid numeric experience data, and the sample may not fully represent the broader Data Analyst job market.

3. Analysis of Programming Language Requirements

The second visualization is a box plot comparing Median Salary Estimate across four programming language categories: Neither, R, Python, and Both. To make the categories consistent, mixed labels such as "R, Python," "Python, R," and "Python/R" were grouped into "Both," while missing or unknown language values were treated as "Neither."

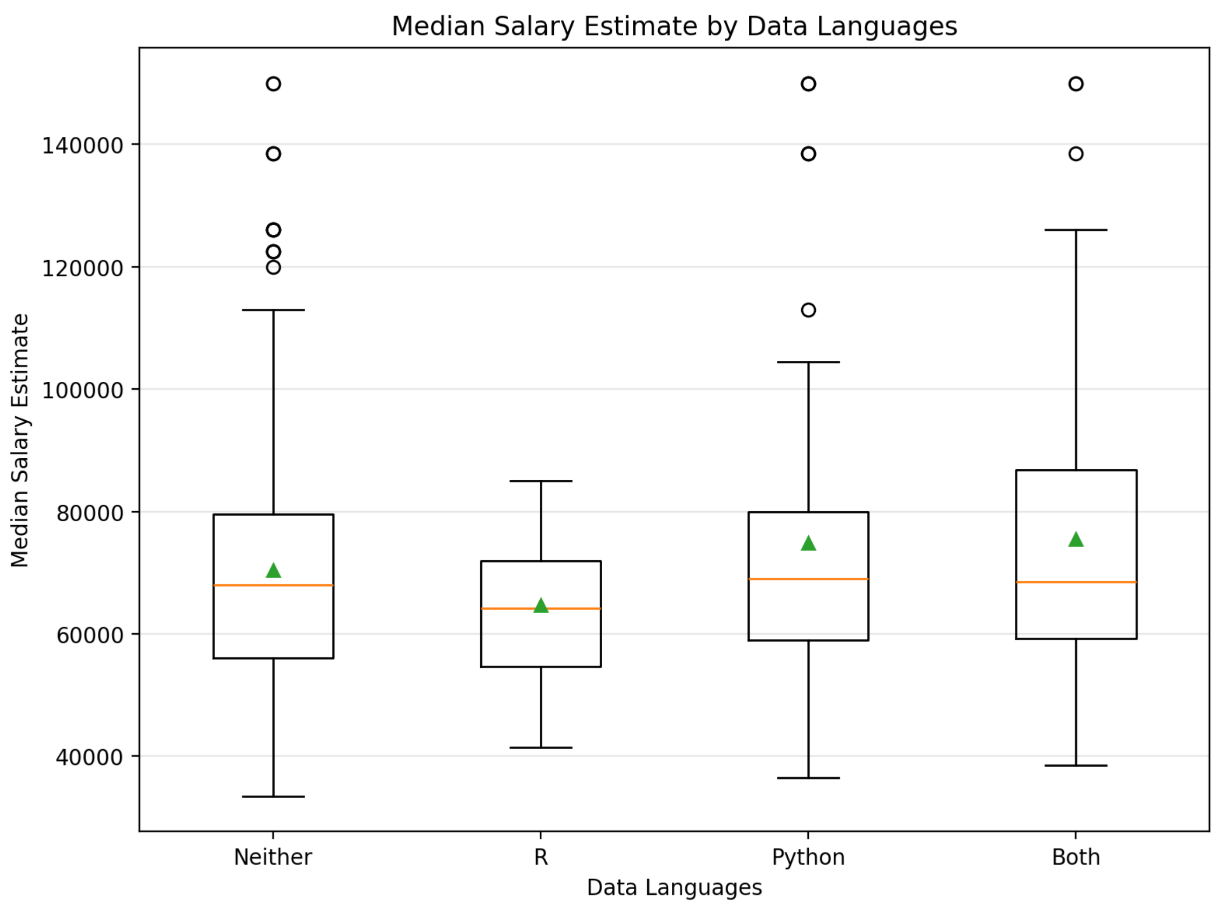

Figure 2. Median Salary Estimate by Data Languages

Interpretation. The box plot suggests that postings requiring Python or both Python and R tend to have slightly higher average salary estimates than postings requiring R only or neither language. However, the medians are relatively close across categories, and there is substantial variation within each group. The "R" category is especially small, with only 14 postings, so its salary distribution should be interpreted cautiously. Overall, the chart suggests that programming language requirements may be associated with salary differences, but the small and uneven category sizes limit the strength of this conclusion.

4. Personal Reflection on the Process

This project helped me understand how data cleaning decisions can shape the final analysis. Some values in the Minimum Years Experience column were missing or non-numeric, so they had to be converted carefully before creating the scatter plot. The Data Languages column also required cleaning because similar categories appeared in different formats, such as “R, Python,” “Python, R,” and “Python/R.”

The process showed that visualizations are useful for quickly identifying patterns, but they must be interpreted with caution. The scatter plot did not show a strong relationship between experience and salary, and the box plot showed some salary differences by programming language category, but the sample size and uneven category distribution limit how strongly these findings can be stated. In future work, I would improve the analysis by using a larger dataset, checking the extraction rules more carefully, and including additional variables such as location, job level, company size, and industry.

Appendix: Key Data Notes

Item	Note
Original sample size	400 job postings
Rows used in scatter plot	311 postings with valid numeric salary and experience values
Rows used in box plot	400 postings with valid salary estimates
Main limitation	Small sample size, uneven language categories, and some extracted values requiring cleaning